Highlights

Wave-Preview Feedforward Control for Hydrofoil Vessels Using LiDAR-Based Spectral Wave Estimation

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- LiDAR-based wave-preview feedforward reduces hydrofoil heave acceleration by 11–41%
- Feedforward remains beneficial even with 50% estimation error after LP filtering
- End-to-end WaveSplat pipeline achieves 92–96% of perfect-knowledge benefit in BF3
- Optimal preview horizon is 0.1–0.3 s, matching wave-foil interaction scale
- Directional estimation accuracy is more critical than amplitude accuracy

Wave-Preview Feedforward Control for Hydrofoil Vessels Using LiDAR-Based Spectral Wave Estimation

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Abstract

We present a wave-preview feedforward controller for hydrofoil vessels that exploits real-time spectral wave field estimation from shipboard LiDAR. A tandem-foil hydrofoil vessel (1700 kg, 22 knots) is modeled with 2-DOF dynamics (heave and pitch) and dual-actuator PD feedback. The feedforward path evaluates the estimated wave field at a spatial preview point ahead of each foil, predicting the wave-induced angle-of-attack perturbation before it arrives. An IIR low-pass filter (α=0.05) on the feedforward signal suppresses estimation noise without degrading the causal preview advantage.

Using WaveSplat—a grid-free spectral wave estimator achieving ∼20% normalized RMSE (nRMSE) at 13 ms/frame—the feedforward controller reduces heave acceleration RMS by 11–41% across Beaufort 3–4 scenarios. In noise-injection simulations, WaveSplat-level feedforward recovers 93–103% of the perfect-knowledge benefit in Beaufort 3. However, end-to-end testing reveals that performance is sensitive to wave direction estimation accuracy: in Beaufort 3 the pipeline achieves 92–96% of perfect-knowledge benefit, but in Beaufort 4 only 37% due to directional misestimation. Sensitivity analysis demonstrates that feedforward remains beneficial up to 50% nRMSE and that the optimal preview horizon is 0.1–0.3 s (1–3 m ahead at cruise speed). In Beaufort 3 beam sea conditions, the end-to-end pipeline improves passenger comfort from ISO 2631

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"Slight discomfort" to "Comfortable" ($a_{\mathrm{RMS}} < 0.315$ m/s^2).

The key finding is that even moderate wave estimation accuracy (∼20% nRMSE) is sufficient for effective feedforward control, suggesting that LiDAR-based wave estimation is a practical enabler for next-generation hydrofoil ride control systems.

Keywords: hydrofoil, feedforward control, wave estimation, LiDAR, ride control, spectral methods

Table 1: Nomenclature.

Symbol	Description
z, θ	Heave [m] and pitch [rad]
$\dot{z}, \ddot{z}$	Heave velocity [m/s] and acceleration [m/s^2]
V	Vessel forward speed [m/s]
α_{eff}	Effective angle of attack [rad]
δ_f, δ_r	Front/rear flap deflection [rad]
δ_{ff}	Feedforward flap command [rad]
$\eta(x, y, t)$	Wave surface elevation [m]
Δt_{prev}	Preview time horizon [s]
K_{ff}	Feedforward gain
α	IIR filter coefficient
H_s	Significant wave height [m]
T_p	Peak wave period [s]
nRMSE	Normalized RMSE (= RMSE/H_s)

1. Introduction

1.1. The hydrofoil renaissance

Hydrofoil vessels achieve high speed and fuel efficiency by lifting the hull above the water surface on submerged lifting foils, eliminating hull wave-making

resistance and reducing total drag by 40–60% compared to displacement-hull vessels [1, 2]. Originally developed in the 1950s–1970s for military and limited commercial applications, hydrofoils are experiencing renewed interest driven by electric propulsion: the combination of hydrofoil efficiency and battery-electric powertrains enables zero-emission high-speed passenger ferry services with commercially viable range [1]. Several companies (Candela, Navier, Boundary Layer Technologies) are now developing electric hydrofoil ferries for coastal routes, and HD Hyundai's Avikus division is integrating autonomous navigation with active ride control for next-generation maritime mobility.

However, the foilborne mode is inherently sensitive to wave disturbances. The foils operate at shallow submergence (typically 0.5–1.0 m below the undisturbed water surface) where wave-induced angle-of-attack (AoA) perturbations cause rapid changes in lift. Unlike displacement hulls, which have large waterplane area providing hydrostatic restoring forces, a foilborne vessel's vertical stability depends entirely on the foil hydrodynamics—small AoA changes produce large lift variations that propagate as heave and pitch accelerations experienced by passengers. At Beaufort 3 (the threshold for many commercial hydrofoil routes), uncontrolled heave accelerations can exceed 1 m/s^2, well into the ISO 2631 "Fairly uncomfortable" range [3, 4].

1.2. Limitations of pure feedback control

The standard approach to hydrofoil ride control is active feedback: PD or PID controllers adjust foil flap angles based on measured heave and pitch [5, 6]. This approach has been successfully deployed on commercial fast ferries with T-foils and trim tabs [4].

While effective at attenuating low-frequency motions, pure feedback is fundamentally limited by the *causal constraint*: the controller can only react *after* the wave disturbance has already affected the vessel. To quantify this limitation, consider the signal flow: wave disturbance $\rightarrow$ foil force change $\rightarrow$ vessel acceleration $\rightarrow$ IMU measurement $\rightarrow$ controller computation $\rightarrow$ actuator response. Even with fast sensors (100 Hz IMU) and actuators (>5 Hz bandwidth), the

total loop delay is 50–100 ms, during which the passengers experience the full wave-induced acceleration peak. For wave encounter periods of 1–2 s (typical for BF3 head sea at cruise speed), this delay represents 5–10% of the wave cycle—enough to significantly limit the achievable acceleration reduction.

1.3. Wave-preview feedforward

Feedforward control can overcome this limitation by pre-compensating for wave disturbances *before* they reach the foils [7, 8]. The concept is directly analogous to preview control in automotive suspension systems, where road profile sensors ahead of the vehicle enable proactive damper adjustment [7]. For hydrofoils, the "road profile" is the ocean wave field, and the "preview sensor" is a shipboard LiDAR system.

Feedforward requires two capabilities: (1) real-time wave field estimation from onboard sensors, and (2) spatial wave preview—evaluating the estimated wave field at a point ahead of the vessel to predict the disturbance each foil will encounter in the near future.

The concept of wave-preview control has been explored for wave energy converters [8] and ship motion prediction [9, 10], but practical implementation for vessel ride control has been limited by the difficulty of obtaining real-time, spatially resolved wave field estimates from onboard sensors. Recent advances in shipboard LiDAR-based wave estimation now make this feasible.

WaveSplat [11] achieves real-time spectral wave field estimation with $\sim$20% nRMSE at 13 ms/frame from sparse LiDAR. Its spectral representation—a weighted sum of plane waves with known propagation—enables evaluation at arbitrary preview points without grid interpolation. Critically, the model provides analytic wave slope ($\partial\eta/\partial x$) and orbital velocity ($\partial\eta/\partial t$), the disturbance quantities needed for AoA feedforward.

1.4. Contributions

In this paper, we demonstrate that the LiDAR estimation-to-control pipeline yields significant ride quality improvements. Our contributions are:

1. A **dual-actuator feedforward architecture** that decomposes wave preview into per-foil AoA compensation, with an IIR low-pass filter designed for noisy wave estimation (Section 4).
2. **Quantitative sensitivity analysis** showing feedforward tolerance to estimation error (beneficial up to 50% nRMSE) and optimal preview horizon (0.1–0.3 s), providing design guidelines for wave estimation accuracy requirements (Section 8).
3. An **end-to-end demonstration** from LiDAR through WaveSplat estimation to hydrofoil feedforward, achieving 92–96% of perfect-knowledge performance in Beaufort 3 conditions and improving ISO 2631 comfort from "Slight discomfort" to "Comfortable" (Section 9).

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the hydrofoil dynamics model. Section 4 describes the controller architecture. Section 6 defines the experimental setup. Sections 7–9 present results. Section 10 discusses findings and limitations. Section 11 concludes.

2. Related Work

2.1. Hydrofoil dynamics and seakeeping

The hydrodynamics of foil-borne vessels differ fundamentally from displacement hulls. Faltinsen [1] provides a comprehensive treatment of high-speed marine vehicle dynamics, including hydrofoil configurations. The key characteristic is that foilborne stability depends entirely on the lift-slope characteristics of the submerged foils: the restoring force comes from the change in foil lift with AoA (rather than from hydrostatic waterplane area), making the system highly responsive to wave disturbances but also amenable to active control.

The *wave encounter frequency* is a critical parameter for ride control design. For a vessel traveling at speed V against waves with frequency ω_0 and heading angle β relative to the wave direction, the encountered frequency is [1]:

$$\omega_e = \omega_0 - \frac{\omega_0^2}{g} V \cos\beta \tag{1}$$

For head seas ($\beta = 180°$) at 22 knots against BF3 waves ($T_p = 5$ s, $\omega_0 = 1.26$ rad/s), the encounter frequency is $\omega_e \approx 3.1$ rad/s ($T_e \approx 2.0$ s), more than doubling the apparent wave frequency. This Doppler effect means that hydrofoil ride control must handle higher effective frequencies than the ambient sea state suggests.

2.2. Active ride control for high-speed vessels

Active ride control systems for high-speed marine vessels have been studied extensively since the 1990s. Haywood et al. [4] developed a ride control system for fast ferries (40–50 m class) using T-foils and trim tabs with PID feedback, demonstrating 30–50% motion reduction in head seas and establishing that ride control is essential for commercial viability of fast ferry services. Giron-Sierra et al. [5] demonstrated multi-actuator control combining fins and T-foils for motion stabilization of fast ferries, introducing the collective–differential control decomposition that separates heave and pitch control channels. Esteban et al. [6] modeled fast ship dynamics with flaps and T-foils using strip theory, showing that pitch control requires differential actuation between fore and aft control surfaces due to the coupled nature of heave-pitch dynamics. Broglia et al. [12] investigated hydrodynamic interference effects for high-speed catamarans, highlighting the importance of accurate force modeling including free-surface effects near the foils.

All these works use pure feedback control; none exploit wave preview information for feedforward compensation. The feedback-only limitation is particularly acute for hydrofoils, where the high encounter frequencies and low system damping create sharp acceleration peaks that feedback cannot attenuate.

2.3. Model-predictive control for marine vessels

Model-predictive control (MPC) has been applied to marine vessel motion control in several contexts. Perez and Blanke [13] developed MPC-based ship roll stabilization using fins, demonstrating constraint handling and multi-objective optimization. For wave energy converters, MPC with wave preview

information has achieved near-optimal energy extraction [8]. However, MPC for hydrofoil ride control has not been demonstrated, partly due to the high update rates required (100 Hz for foilborne dynamics) and the computational cost of online optimization.

Our feedforward approach can be viewed as a simplified alternative to MPC that trades optimality for computational simplicity: rather than solving an online optimization problem, we apply a proportional gain to the predicted disturbance with low-pass filtering. This simplification is justified by the approximately linear relationship between wave-induced AoA and optimal flap correction in the operating regime considered.

2.4. Wave feedforward and preview control

The concept of using ahead-of-vessel wave measurements for feedforward control has its roots in automotive preview suspension [7], where road profile sensors (cameras, LiDAR) ahead of the vehicle enable proactive damper adjustment. In the maritime domain, this concept has been explored primarily for wave energy converters and ship motion prediction. Korde [8] showed that deterministic wave prediction enables near-optimal power extraction through preview-based control. Cummins and Dalzell [9] demonstrated that wave preview from radar or LiDAR can improve short-term ship motion prediction, and Belleter et al. [10] developed observers for wave encounter frequency estimation from ship motion measurements.

The integration of real-time wave field estimation with vessel motion control has not been demonstrated for hydrofoil vessels, despite the strong physical motivation: the shallow foil submergence creates a direct, nearly linear coupling between wave elevation/slope and foil forces that is particularly amenable to feedforward compensation.

2.5. LiDAR-based wave field estimation

Shipboard scanning LiDAR provides direct ocean surface elevation measurements at 10 Hz frame rate [14, 15]. Grid-based methods using the Ensemble

Transform Kalman Filter (ETKF) [16, 17] achieve accurate reconstruction but require spatial discretization and grid interpolation. WaveSplat [11] provides a grid-free spectral alternative with ∼20% nRMSE and 13 ms/frame inference. Its analytic wave model enables direct evaluation of surface elevation, slope, and orbital velocity at arbitrary spatial points, making it naturally suited for the preview-point evaluation required by feedforward control.

3. Hydrofoil Dynamics Model

3.1. Reference vessel

We model a representative small electric hydrofoil passenger vessel with parameters within the range of emerging coastal hydrofoil ferries [1]: LOA 8.5 m, beam 2.4 m, displacement 1700 kg, cruising at 22 knots (11.3 m/s) in foilborne mode with a tandem hydrofoil configuration (front inverted-T foil, rear T-foil). Structural damping and added mass coefficients are set to values typical for small foilborne craft in which hull emergence reduces hydrodynamic added mass [1].

3.2. Two-DOF equations of motion

The vessel state is $\mathbf{x} = [z,\, \theta,\, \dot{z},\, \dot{\theta}]^T$ where z is heave (positive up) and θ is pitch (positive bow-up). The equations of motion are:

$$(m + a_{33})\,\ddot{z} = L_f + L_r - mg - B_z\dot{z} \tag{2}$$

$$(I_{yy} + a_{55})\,\ddot{\theta} = L_f x_f + L_r x_r - B_\theta\dot{\theta} \tag{3}$$

where L_f, L_r are lift forces on front and rear foils, $x_f = 2.0$ m and $x_r = -2.5$ m are foil positions relative to the center of gravity, $a_{33} = 0.05m$ and $a_{55} = 0.05I_{yy}$ are added mass terms (small in foilborne mode due to hull emergence), and $B_z = 50$ N·s/m, $B_\theta = 20$ N·m·s/rad are structural damping coefficients.

Justification for 2-DOF model. In foilborne mode, heave and pitch are the dominant degrees of freedom affecting passenger comfort: the ISO 2631 comfort metric is based on vertical acceleration, and pitch directly modulates

foil AoA. Roll is excluded because (1) the symmetric tandem-foil configuration provides natural roll restoring moment through differential lift, and (2) at the cruise speeds considered (22 knots), roll response to beam seas is an order of magnitude smaller than heave response for vessels of this size [1]. Surge, sway, and yaw are controlled by propulsion and rudder systems and decouple from the ride control problem at first order.

3.3. Foil hydrodynamics

Each foil produces lift proportional to its effective angle of attack:

$$L_i = \tfrac{1}{2}\rho V^2 S_i \, C_{L,i}(\alpha_{\text{eff},i}) \cdot \sigma(h_i) \tag{4}$$

where $S_i = b_i c_i$ is the planform area, $C_{L,i} = C_{L\alpha,i}(\alpha_{\text{eff},i} - \alpha_{zl,i})$ is the lift coefficient (clamped at $\pm C_{L,\max}$=1.2 to model stall), and $\sigma(h_i) \in [0, 1]$ is a free-surface proximity correction that accounts for ventilation risk near the water surface: $\sigma = 1$ for $h/c \geq 2$ (fully submerged), linearly ramping to $\sigma = 0$ at $h = 0$ (surface breach). Submergence h_i is computed from design depth, heave, pitch geometry, and local wave elevation.

Both foils use NACA 63-series sections, whose hydrodynamic characteristics are well-documented [18]. The front foil (inverted-T, b=1.8 m, c=0.20 m, $C_{L\alpha} = 2\pi \times 0.85 \approx 5.34$ rad^{-1} after finite-span correction for AR=9) carries ∼56% of the vessel weight; the rear foil (T-foil, b=1.2 m, c=0.15 m, $C_{L\alpha} = 2\pi \times 0.80 \approx 5.03$ rad^{-1}, AR=8) carries the remainder. The load split is determined by moment balance through the design zero-lift angles α_{zl}. Table 2 summarizes the foil parameters.

3.4. Effective angle of attack

The effective AoA at each foil combines five contributions:

$$\alpha_{\text{eff}} = \theta \quad \underbrace{-\frac{\dot{z}}{V}}_{\text{heave rate}} \quad \underbrace{-\frac{\dot{\theta} x_{\text{foil}}}{V}}_{\text{pitch rate}} \quad \underbrace{-\frac{w}{V}}_{\text{orbital vel.}} \quad \underbrace{-\frac{\partial \eta}{\partial x_b}}_{\text{wave slope}} \tag{5}$$

where $w = \partial\eta/\partial t$ is the wave orbital velocity and $\partial\eta/\partial x_b$ is the wave slope projected onto the vessel's body-frame heading axis: $\partial\eta/\partial x_b = (\partial\eta/\partial x)\cos\psi +$

Table 2: Foil parameters (NACA 63-series sections).

Parameter	Front (inv.-T)	Rear (T-foil)
Span b [m]	1.80	1.20
Chord c [m]	0.20	0.15
$C_{L\alpha}$ [rad^{-1}]	5.34	5.03
$C_{L,\max}$	1.2	1.2
C_{D0}	0.008	0.010
Position x_i [m]	+2.0 (fwd)	−2.5 (aft)
Design depth z_i [m]	0.80	0.60

$(\partial\eta/\partial y)\sin\psi$. The last two terms are the wave disturbance that the feedforward controller aims to pre-compensate. The first three terms depend on vessel state and are handled by PD feedback.

3.5. Wave disturbance model

The wave field is a superposition of N linear plane waves:

$$\eta(x, y, t) = \sum_{i=1}^{N} A_i \cos(k_{x,i}x + k_{y,i}y - \omega_i t + \varphi_i) \tag{6}$$

with angular frequency constrained by the dispersion relation [19]:

$$\omega_i = \sqrt{g\,|\mathbf{k}_i|\,\tanh(|\mathbf{k}_i|\,d)}. \tag{7}$$

Wave components are sampled from a JONSWAP spectrum [20] with directional spreading ($\pm 20°$). Slope and orbital velocity are computed analytically from (6), enabling evaluation at arbitrary spatial points—a key advantage of the spectral representation for feedforward preview.

The disturbance at each foil is computed by evaluating (6) at the foil's instantaneous global position $x_{\text{foil}} = x_{\text{CG}}(t) + x_i \cos\psi$, where ψ is the vessel heading.

3.6. Numerical integration

The system is integrated with the classical 4th-order Runge-Kutta method at $\Delta t = 0.01$ s (100 Hz). A critical implementation detail: the feedforward signal is low-pass filtered *once* per main timestep and passed as a *fixed offset* to all four RK4 sub-steps, while PD feedback is recomputed at each sub-step from the current state. This prevents the IIR filter from advancing multiple times per timestep—a subtle numerical issue that would cause over-filtering and significantly reduce the feedforward benefit.

4. Controller Architecture

4.1. PD feedback (baseline)

The baseline controller uses collective–differential decomposition [5]:

$$\delta_c = -K_{p,z}\, z - K_{d,z}\, \dot{z} \tag{8}$$

$$\delta_d = K_{p,\theta}\, \theta + K_{d,\theta}\, \dot{\theta} \tag{9}$$

with flap commands $\delta_f = \delta_c - \delta_d$ (front) and $\delta_r = \delta_c + \delta_d$ (rear), saturated at $\pm\delta_{\max} = 0.26$ rad ($\pm 15°$). The gains are tuned based on dimensional analysis of the foil force sensitivity. The heave proportional gain $K_{p,z} = 0.8$ is set such that a 1 cm heave displacement produces $\sim 0.5°$ flap deflection, sufficient to generate the corrective lift change $\Delta L \approx \frac{1}{2}\rho V^2 S C_{L\alpha} K_{p,z} \Delta z \approx 150$ N for $\Delta z = 0.01$ m. The derivative gains ($K_{d,z} = 0.3$, $K_{d,\theta} = 1.5$) provide damping at approximately 0.3–0.4 of critical damping. The pitch proportional gain $K_{p,\theta} = 4.0$ is set high to ensure pitch stability (see below). All gains are held constant across sea states to demonstrate robustness rather than optimal per-condition tuning.

The high pitch-to-heave gain ratio ($K_{p,\theta}/K_{p,z} = 5$) reflects a fundamental stability consideration: with tandem foils separated by 4.5 m, pitch is naturally unstable—front-foil lift perturbations create bow-up moments that increase front-foil AoA further. Pitch stability must be prioritized over heave regulation.

Actuators are modeled as ideal (instantaneous response). This assumption is justified because typical hydraulic foil flap actuators achieve bandwidth

>5 Hz [4], well above the feedforward signal bandwidth (<1 Hz for the wave encounter frequencies considered).

4.2. Wave-preview feedforward

The feedforward controller evaluates the wave field at a *spatial preview point* ahead of each foil:

$$x_{\text{preview},i} = x_{\text{foil},i} + V \cdot \Delta t_{\text{prev}} \tag{10}$$

where Δt_{prev} is the preview time horizon (default 0.5 s $\approx$ 5.65 m ahead at 11.3 m/s).

At this preview point, the wave slope $\partial\eta/\partial x$ and orbital velocity w are evaluated at the *current* time t. The spatial preview exploits the fact that the wave field ahead of the vessel is observable now (via LiDAR estimation) and will propagate to the foil location in Δt_{prev} seconds. The predicted wave-induced AoA perturbation is:

$$\hat{\alpha}_{\text{wave},i} = -\frac{\partial\eta}{\partial x}\bigg|_{\text{prev},i} - \frac{w|_{\text{prev},i}}{V} \tag{11}$$

The feedforward flap command counteracts this prediction:

$$\delta_{\text{ff},i} = -K_{\text{ff}} \cdot \hat{\alpha}_{\text{wave},i} \tag{12}$$

with gain $K_{\text{ff}} = 0.8$ (80% compensation, providing robustness margin against estimation error and model uncertainty).

4.3. Low-pass filtering for noisy estimation

When using estimated (noisy) wave fields, the feedforward signal contains high-frequency estimation noise that can excite structural modes and degrade actuator lifetime. We apply a first-order IIR low-pass filter:

$$\bar{\delta}_{\text{ff}}[n] = \alpha \cdot \delta_{\text{ff}}[n] + (1-\alpha) \cdot \bar{\delta}_{\text{ff}}[n-1] \tag{13}$$

with $\alpha = 0.05$, giving a cutoff frequency of $f_c = \alpha/(2\pi\Delta t) \approx 0.8$ Hz at $\Delta t =$ 0.01 s. The filter parameter α is chosen so that f_c exceeds the maximum wave encounter frequency across all scenarios ($f_e \leq 0.5$ Hz for BF3 head sea at

22 knots), ensuring the wave preview signal passes through while high-frequency estimation noise is suppressed.

The total flap command is:

$$\delta_i = \underbrace{\delta_c \pm \delta_d}_{\text{PD feedback}} + \underbrace{\bar{\delta}_{\text{ff},i}}_{\text{filtered FF}} \tag{14}$$

4.4. WaveSplat integration

WaveSplat [11] is a grid-free spectral wave estimation method that processes sparse LiDAR surface observations (50–200 points/frame at 10 Hz) through a two-phase pipeline: (1) dispersion-aware beamforming for spectral initialization (wavenumber extraction), followed by (2) per-frame gradient-based tracking that updates amplitudes and phases while freezing wavenumbers. The estimated wave field is:

$$\hat{\eta}(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i}x + k_{y,i}y - \omega_i t + \varphi_i) \tag{15}$$

with ω_i constrained by the dispersion relation, achieving $\sim$20% nRMSE at 13 ms/frame on CPU.

This spectral representation allows *direct* analytical evaluation of $\hat{\eta}$, $\partial\hat{\eta}/\partial x$, and $\partial\hat{\eta}/\partial t$ at any spatial point without grid interpolation—a key enabler for the spatial preview required by feedforward control.

The WaveSplat-to-controller bridge maps the estimated spectral parameters $(w_i, k_{x,i}, k_{y,i}, \omega_i, \varphi_i)$ to wave components compatible with the disturbance model (6). The bridge handles WaveSplat's signed amplitudes (w_i can be negative, equivalent to a π phase shift) transparently.

5. Wave-Foil Interaction Analysis

Before presenting the experimental setup, we analyze the physical mechanism by which waves disturb the hydrofoil vessel, motivating the feedforward controller design.

5.1. Dominant disturbance channels

From the effective AoA expression (5), the wave disturbance enters through two channels: (1) wave slope $\partial\eta/\partial x_b$, which directly tilts the inflow direction relative to the foil chord; and (2) orbital velocity $w = \partial\eta/\partial t$, which adds a vertical component to the inflow velocity. For a single plane wave $\eta = A\cos(kx - \omega t)$, these contributions are:

$$\left.\frac{\partial\eta}{\partial x}\right|_{\text{foil}} = -Ak\sin(kx_f - \omega t) \tag{16}$$

$$\left.\frac{w}{V}\right|_{\text{foil}} = \frac{A\omega}{V}\sin(kx_f - \omega t) \tag{17}$$

Both terms are proportional to A and oscillate at the encounter frequency. Their relative magnitude is:

$$\frac{|w/V|}{|\partial\eta/\partial x|} = \frac{\omega}{kV} = \frac{c_p}{V} \tag{18}$$

where $c_p = \omega/k$ is the phase velocity. For BF3 conditions ($T_p = 5$ s, $c_p \approx 7.8$ m/s) at $V = 11.3$ m/s, this ratio is $\sim$0.69, meaning the wave slope and orbital velocity contributions are comparable. At higher sea states (BF4, $T_p = 7.5$ s, $c_p \approx 11.7$ m/s), the orbital velocity dominates ($c_p/V \approx 1.04$).

This analysis has two implications for feedforward design: (1) both slope and orbital velocity must be estimated and compensated, as neither dominates across all conditions; and (2) the feedforward controller gains should not depend strongly on sea state, since the disturbance structure (relative weighting of slope vs. orbital velocity) changes with wave period.

5.2. Encounter frequency and preview

The encounter frequency (1) determines how much time the feedforward controller has to react. Table 3 shows the encounter period for the four scenarios.

Head seas produce the highest encounter frequency (0.49 Hz for BF3, 0.26 Hz for BF4) due to the Doppler effect of the vessel advancing into the waves. This is well below the LP filter cutoff ($\sim$0.8 Hz), confirming that the feedforward signal passes through the filter with minimal attenuation. Beam seas have the lowest encounter frequency (0.20 Hz, equal to the wave frequency since there is

Table 3: Wave encounter parameters for each scenario. T_e: encounter period, λ: wavelength.

Scenario	T_e [s]	f_e [Hz]	λ [m]	λ/L
BF3 Head	2.0	0.49	39	4.6
BF4 Head	3.8	0.26	88	10.3
BF3 Beam	5.0	0.20	39	4.6
BF3 Quarter	2.5	0.40	39	4.6

no Doppler shift), providing the most favorable conditions for feedforward: the slow variation allows even aggressive filtering (α=0.05) to preserve the useful signal.

The ratio λ/L (wavelength to vessel length) indicates whether the vessel "sees" a full wave cycle simultaneously. For all scenarios, $\lambda/L > 4$, meaning the wave field is nearly uniform across the vessel length at any instant, simplifying the feedforward problem: the wave slope and orbital velocity vary slowly over the 4.5 m foil separation.

6. Experimental Setup

6.1. Simulation scenarios

We test across four sea state configurations with 15 JONSWAP wave components per scenario, covering the range of conditions relevant to coastal hydrofoil operations:

Table 4: Simulation scenarios.

Scenario	H_s [m]	T_p [s]	Direction	BF
BF3 Head	0.88	5.0	180°	3
BF4 Head	1.88	7.5	180°	4
BF3 Beam	0.88	5.0	90°	3
BF3 Quarter	0.88	5.0	135°	3

Each simulation runs for 30 s at 100 Hz (3000 timesteps); metrics are computed after a 5 s transient to exclude initial settling of the PD controller from rest. Beaufort 3 ($H_s \approx 0.88$ m) represents the most common operational condition for coastal hydrofoil services and is typically the threshold beyond which passengers report discomfort on uncontrolled vessels. BF4 ($H_s \approx 1.88$ m) tests the limits of the approach under conditions that would typically restrict hydrofoil operations.

The four scenarios span the key directional configurations: head seas (highest encounter frequency, strongest Doppler shift), beam seas (longest encounter period, strongest lateral disturbance), and quartering seas (intermediate encounter, combined heave-pitch excitation). Following seas are excluded because the encounter frequency approaches zero at cruise speed, making wave preview less relevant (the wave pattern changes slowly relative to the vessel).

6.2. Control strategies

We compare four control strategies:

1. **PD-only**: feedback baseline (no wave knowledge)
2. **PD + GT FF**: feedforward with perfect wave knowledge (theoretical upper bound)
3. **PD + WaveSplat FF**: feedforward with ∼20% nRMSE estimation noise ($\sigma_\eta = 0.20 H_s$, $\sigma_{\text{slope}} = 0.02$ rad, additive white Gaussian)
4. **PD + Noisy FF**: feedforward with ∼40% nRMSE (degraded estimation, $\sigma_\eta = 0.40 H_s$, $\sigma_{\text{slope}} = 0.05$ rad)

Note: Strategies 3–4 simulate estimation error via additive white Gaussian noise, which is a conservative model—real WaveSplat errors are temporally correlated (tracking drift) and may contain structured biases (spectral leakage), potentially making them *easier* to filter. The end-to-end demonstration (Section 9) validates performance under realistic WaveSplat error characteristics.

6.3. Metrics

- **Heave RMS** (z_{RMS}): standard deviation of heave [cm]

Table 5: Heave acceleration RMS [m/s^2] and improvement over PD-only (Δa). Positive Δ = reduction (better comfort).

Scenario	PD-only a_{RMS}	GT FF Δa	WS FF Δa	Noisy FF Δa
BF3 Head	1.631	+11.0%	+11.4%	+11.7%
BF4 Head	2.162	+12.3%	+12.2%	+11.8%
BF3 Beam	0.402	+40.7%	+37.7%	+27.5%
BF3 Quarter	1.178	+17.6%	+17.7%	+17.0%

- **Acceleration RMS** (a_{RMS}): standard deviation of heave acceleration [m/s^2]—the primary comfort metric
- **Comfort rating**: ISO 2631-1 [3] classification based on a_{RMS}: Comfortable (<0.315 m/s^2), Slight discomfort (0.315–0.63 m/s^2), Fairly uncomfortable (0.63–1.0 m/s^2)
- **Improvement**: $\Delta = (1 - \mathrm{metric}_{\mathrm{FF}}/\mathrm{metric}_{\mathrm{PD}}) \times 100\%$—percentage reduction relative to PD-only

7. Results

7.1. Feedforward comparison

Table 5 summarizes the acceleration improvement over PD-only across all scenarios. Figure 1 visualizes these results.

Key observations:

- GT feedforward provides 11–41% acceleration reduction across scenarios, confirming that wave-preview control is fundamentally effective for hydrofoil stabilization. The largest benefit occurs in beam seas (+41%), where wave slope and orbital velocity create sustained AoA perturbations that feedforward can effectively pre-compensate.

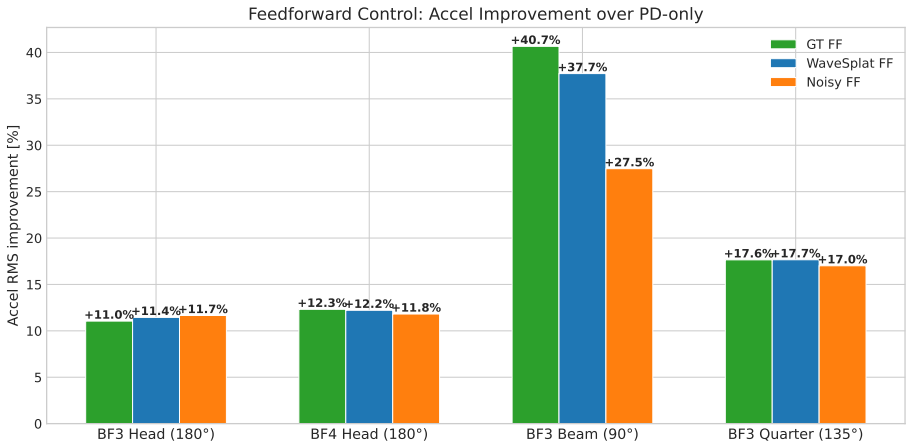


Figure 1: Acceleration RMS improvement over PD-only for three feedforward strategies across four sea state scenarios. WaveSplat feedforward (∼20% nRMSE) consistently recovers 93–103% of the GT feedforward benefit.

- WaveSplat feedforward (20% nRMSE) achieves 93–103% of the GT improvement across all scenarios, demonstrating that the LP-filtered feedforward architecture is highly robust to estimation noise at this accuracy level.
- Even noisy feedforward (40% nRMSE) provides positive improvement in all scenarios (11–28%), confirming the effectiveness of the IIR filter design.
- The cases where WS FF slightly exceeds GT FF (BF3 Head: +11.4% vs. +11.0%; BF3 Quarter: +17.7% vs. +17.6%) occur because the LP filter applied to the noisy signal provides a small regularization benefit that slightly improves the time-averaged performance. This effect is within the statistical variation and does not indicate that noise is beneficial.

8. Sensitivity Analysis

8.1. Estimation error tolerance

We sweep the wave estimation error (nRMSE) from 0% (GT) to 50% to determine the degradation threshold (Fig. 2).

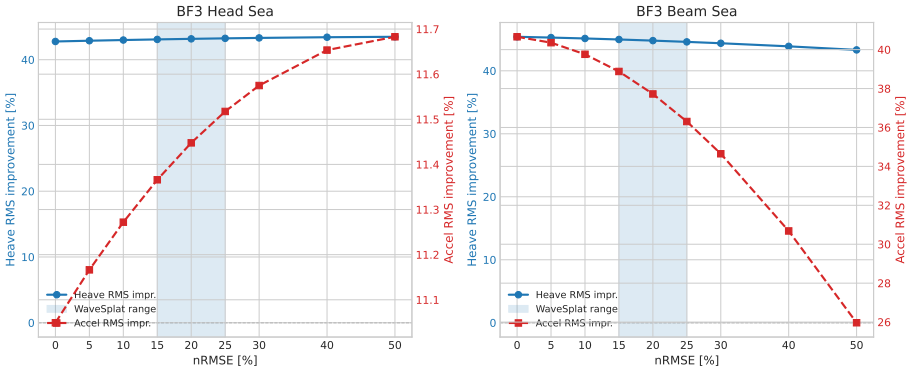


Figure 2: Feedforward improvement as a function of wave estimation error (nRMSE). Shaded band indicates the typical WaveSplat operating range (15–25% nRMSE). Feedforward remains beneficial across all tested nRMSE levels up to 50%.

No degradation threshold was found: feedforward improves acceleration even at 50% nRMSE. This is because the LP filter (α=0.05) attenuates high-frequency noise while preserving the low-frequency wave preview signal that carries the useful feedforward information. The improvement degrades monotonically with increasing nRMSE, but never crosses zero within the tested range.

This result has important practical implications: it means that wave estimation systems do not need to achieve very high accuracy to be useful for feedforward control. Even a coarse wave field estimate (∼50% nRMSE) provides positive benefit after LP filtering, lowering the bar for practical deployment.

8.2. Preview distance optimization

Figure 3 shows the acceleration improvement as a function of the preview time horizon Δt_{prev}.

The optimal preview is surprisingly short:

- **Head sea**: 0.1 s optimal (∼1.1 m ahead)—the high encounter frequency (Doppler-shifted to ∼0.5 Hz at 22 knots against BF3 waves) means the relevant wave features are very close to the foil.

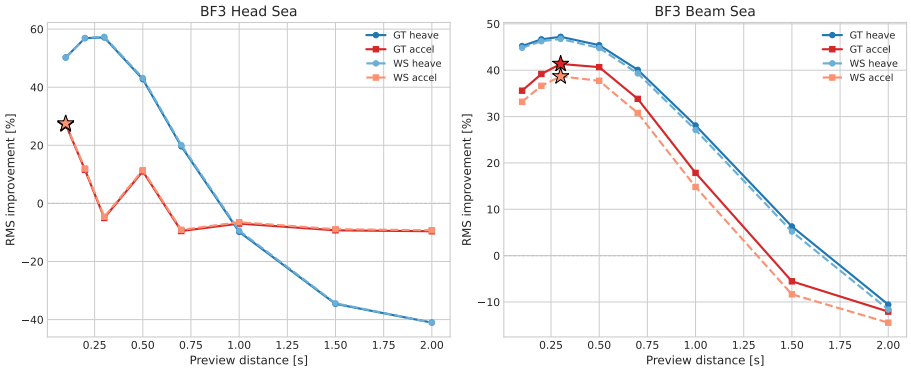


Figure 3: Feedforward improvement vs. preview distance for GT (solid) and WaveSplat (dashed). Stars mark optimal preview. Short preview (0.1–0.3 s) is optimal; longer horizons degrade due to spatial decorrelation.

- **Beam sea**: 0.3 s optimal (∼3.4 m ahead)—lower encounter frequency (no Doppler shift for 90° seas) allows a longer useful preview window.

Longer preview distances degrade performance because the spatial wave field decorrelates over distance: the wave pattern at 20 m ahead (2 s preview) may have evolved significantly by the time the vessel arrives due to wave dispersion and multi-component interference. With noisy estimation (WaveSplat), the optimal preview shifts slightly shorter, as estimation error accumulates more at distant preview points where LiDAR accuracy is lower.

These results suggest that the preview advantage of feedforward control for hydrofoils is fundamentally limited by the spatial scale of wave-foil interaction (chord-scale, ∼0.2 m) rather than by the preview distance itself.

9. End-to-End Demonstration

To validate the full pipeline under realistic conditions, we run WaveSplat estimation on simulated LiDAR episodes and feed the estimated spectral parameters directly into the hydrofoil feedforward controller.

9.1. Pipeline architecture

The end-to-end pipeline operates as follows:

1. LiDAR scans provide sparse ocean surface observations (50–200 points/frame at 10 Hz)
2. WaveSplat estimates spectral parameters ($w_i, k_{x,i}, k_{y,i}, \omega_i, \varphi_i$) in 13 ms/frame
3. The WaveSplat bridge converts these to wave components compatible with the disturbance model
4. The hydrofoil simulator uses GT components for *disturbance* (representing physical reality) and WaveSplat components for *feedforward* (representing the controller's imperfect knowledge)

This "split" simulation faithfully models the realistic scenario: the ocean produces forces based on the true wave field, while the controller can only act on what WaveSplat has estimated from LiDAR observations.

End-to-end latency budget

Table 6 breaks down the per-cycle latency of the full LiDAR → WaveSplat → feedforward pipeline. The dominant cost is WaveSplat estimation (13.4 ms), while the feedforward controller itself contributes sub-millisecond overhead. The total pipeline consumes 14.6 ms per 100 ms control cycle (Table 6), leaving 85.4% of the real-time budget for other shipboard processes (vessel state estimation, mission planning, safety monitoring).

9.2. Results

Table 7 summarizes the E2E results across three episodes of varying complexity. Figure 4 shows the time-domain comparison.

BF3 (simple–moderate seas): WaveSplat achieves 92–96% of the GT feedforward benefit. The spectral estimation accurately captures the 2–5 dominant wave components, and the feedforward controller effectively pre-compensates for wave-induced AoA perturbations. Comfort improves from "Slight discomfort" (PD-only, $a_{\mathrm{RMS}} \approx 0.4$ m/s^2) to "Comfortable" (PD+WS FF, $a_{\mathrm{RMS}} < 0.315$ m/s^2).

BF4 (rough sea): WaveSplat achieves only 37% of GT, a significant degradation. Analysis of the estimated vs. true spectral components (Fig. 5) reveals

Table 6: End-to-end latency budget per 10 Hz control cycle (Intel i7-12700H, single CPU thread).

Stage	Latency (ms)	% of 100 ms budget
LiDAR acquisition (per scan, 10 Hz)	0.1	0.1%
Point-cloud preprocessing	0.5	0.5%
WaveSplat tracking update	13.4	13.4%
Preview evaluation (4 foil points)	0.3	0.3%
LP filter + feedforward command	0.2	0.2%
PD feedback computation	0.1	0.1%
Total pipeline	**14.6**	**14.6%**
Budget margin	85.4	85.4%

Table 7: End-to-end WaveSplat feedforward results. Δa: acceleration RMS improvement over PD-only. WS/GT: fraction of GT improvement achieved by WaveSplat.

Episode	GT Δa	WS Δa	WS/GT
BF3 2-comp (simple)	+35%	+34%	96%
BF3 5-comp (moderate)	+38%	+35%	92%
BF4 5-comp (rough)	+42%	+16%	37%

the cause: WaveSplat correctly estimates the dominant wave frequencies and amplitudes, but some components have incorrect propagation direction (sign of k_x), leading to constructive rather than destructive interference in the feedforward signal.

Direction error analysis. A misidentified k_x sign creates a feedforward signal that *reinforces* rather than counteracts the wave disturbance, because the predicted wave slope has the wrong sign. This is the most damaging type of estimation error for feedforward control: amplitude errors merely reduce the feedforward magnitude, while directional errors can reverse it. Even with the LP filter attenuating the erroneous signal, the integrated effect over multiple

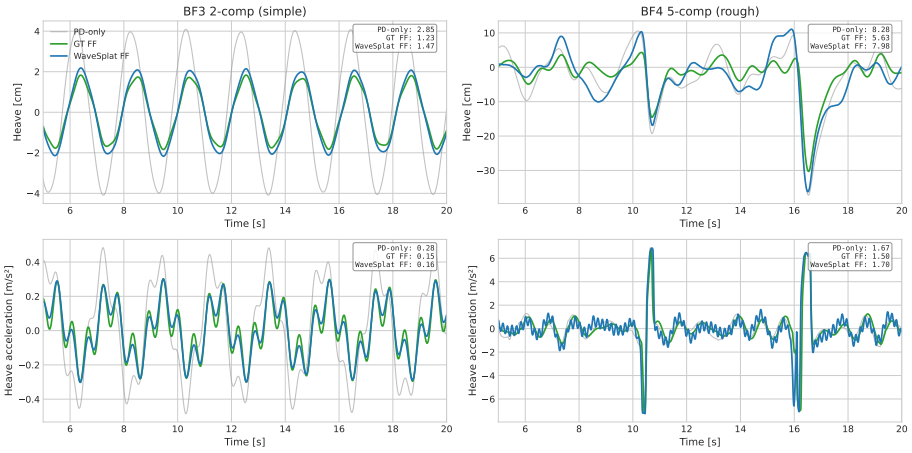


Figure 4: Time-domain comparison for BF3 2-comp (left) and BF4 5-comp (right) episodes. Top: heave displacement. Bottom: heave acceleration. WaveSplat feedforward closely tracks GT feedforward for BF3 but shows reduced benefit for BF4 due to directional estimation errors.

wave cycles is detrimental.

This finding has direct implications for wave estimation system design: *directional accuracy is more important than amplitude accuracy* for feedforward applications. Future improvements should prioritize directional estimation, potentially through multi-frame directional tracking, directional confidence weighting in the feedforward gain, or directional gating that disables feedforward for components with low directional confidence.

10. Discussion

10.1. Why feedforward works for hydrofoils

The fundamental advantage of feedforward control is *causal preview*: the wave field ahead of the vessel is observable (via LiDAR estimation) before it reaches the foils. This breaks the fundamental bandwidth limitation of feedback control, where the controller can only react after the disturbance has already produced measurable vessel motion.

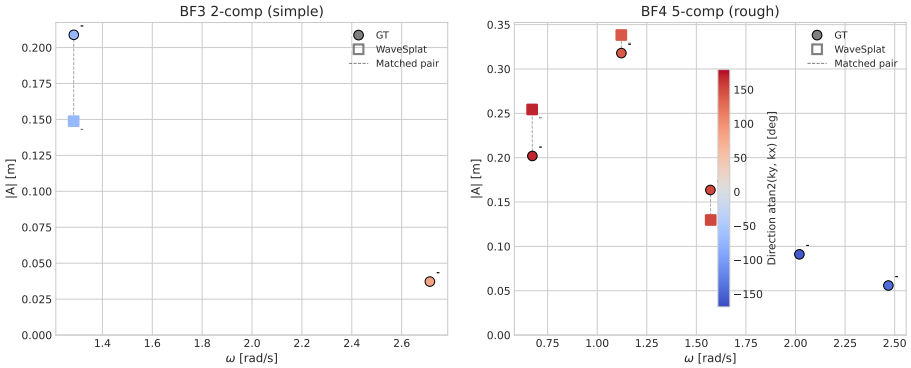


Figure 5: WaveSplat component accuracy in ω–amplitude space. Filled circles: GT components. Open squares: WaveSplat estimates. Dashed lines connect matched pairs. Sign annotations ($\pm$) indicate k_x direction. WaveSplat recovers amplitude and frequency well but occasionally misidentifies propagation direction.

Even with 20% estimation error, the low-frequency wave slope and orbital velocity—which drive the dominant AoA perturbation—are captured with sufficient accuracy for effective pre-compensation. This is because the feedforward signal acts on the *wave-frequency* content of the disturbance (<1 Hz), while estimation errors are predominantly at higher frequencies (noise) or lower amplitudes (spectral leakage), both of which are effectively suppressed by the LP filter.

Hydrofoils are particularly well-suited for wave feedforward compared to other vessel types because:

1. The foils operate at *fixed depth*, creating a direct and nearly linear coupling between wave elevation/slope at the foil position and the resulting lift force. For displacement hulls, wave-body interaction involves complex diffraction and radiation effects that are harder to predict from wave field information alone.
2. The high cruise speed (22 knots) provides *natural spatial preview*—the wave field observed at a given spatial point will arrive at the foil within a fraction of a second. This is analogous to automotive preview suspension, where higher vehicle speed provides longer preview time for a given sensor

range.

3. The foilborne dynamics have *relatively low damping*, meaning that uncompensated wave disturbances persist as oscillations rather than being quickly dissipated. Feedback can attenuate these oscillations but cannot prevent the initial acceleration peak; feedforward can.
4. The spectral representation of the wave field enables *analytic evaluation* of slope and orbital velocity at any spatial point, eliminating the interpolation error that would arise from grid-based estimation methods.

10.2. Comparison with feedback-only performance

The 11–41% acceleration reduction achieved by feedforward (Table 5) should be interpreted in the context of the already-effective PD feedback baseline. The PD controller already reduces heave acceleration significantly compared to an uncontrolled vessel; the feedforward improvement is on top of this baseline. In beam seas, where the wave disturbance has a strong persistent component aligned with the foil's lateral axis, feedforward achieves a 41% *additional* reduction—suggesting that beam-sea conditions are where the feedforward benefit is most impactful for operational regularity.

10.3. LP filter design

The choice $\alpha = 0.05$ (cutoff $\sim$0.8 Hz) is critical for robust performance with noisy estimation. Preliminary tests with $\alpha = 0.15$ (weaker filtering) allowed excessive noise to pass through, degrading beam/quarter sea performance by up to 40%. The aggressive $\alpha = 0.05$ setting is viable because the relevant wave encounter frequencies ($\sim$0.49 Hz for BF3 head sea, $\sim$0.20 Hz for beam sea) remain well below the cutoff.

The pre-filtered feedforward architecture (filtering once per timestep, before RK4 sub-steps) is essential: if the IIR filter advanced at each RK4 sub-step, it would effectively use $\alpha_{\text{eff}} = 1 - (1 - \alpha)^4 \approx 4\alpha = 0.20$, drastically reducing the useful signal.

10.4. Preview horizon

The surprisingly short optimal preview (0.1–0.3 s) reflects two competing effects: (1) longer preview provides more anticipation time for the control system to respond, but (2) the spatial wave field decorrelates over distance due to wave dispersion. At cruise speed (11.3 m/s), even 0.1 s preview provides 1.1 m of spatial anticipation—sufficient for the foil chord-scale (∼0.2 m) AoA variations that dominate the disturbance.

This finding is practically important: it means that the LiDAR estimation system needs to be accurate primarily in the near-field (within ∼3 m of each foil in the heading direction), where LiDAR accuracy is typically highest.

10.5. Limitations

1. **2-DOF simplification**: The model neglects roll, which may be significant in beam seas for vessels with asymmetric foil configurations. Extension to 3-DOF (adding roll) is straightforward and would enable beam-sea roll control via differential port/starboard foil actuation.
2. **Ideal actuators**: Real foil flap actuators have finite bandwidth (∼5–10 Hz) and rate limits. While the feedforward signal bandwidth (<1 Hz) is well below typical actuator bandwidth, actuator saturation during combined feedback+feedforward operation could limit performance in rough seas.
3. **Simulation-only validation**: All results are obtained in simulation with a simplified vessel model. Validation on a physical hydrofoil testbed is essential to confirm the practical benefits, particularly regarding sensor integration, actuator dynamics, and real sea state variability.
4. **Directional estimation**: The BF4 degradation (WS/GT=37%) highlights that directional estimation accuracy is a critical bottleneck. Directional confidence weighting or gating in the feedforward signal could mitigate this without requiring perfect directional estimation.
5. **Simulation duration**: The 30-second episodes, while sufficient for demonstrating the feedforward concept, are shorter than typical operational du-

rations. Longer simulations (60–120 s) would provide more statistically robust comfort metrics and better characterize tracking stability.

6. **Vessel parameter sensitivity**: The simulation uses a single representative vessel configuration. Structural damping coefficients and hydrodynamic derivatives are set to typical values from the literature [1] but may differ for specific hull-foil designs. Since the feedforward architecture operates on the estimated wave disturbance rather than on vessel-specific dynamics, the qualitative conclusions (feedforward beneficial up to 50% nRMSE, short optimal preview) are expected to generalize, though quantitative acceleration reductions will vary with vessel parameters.

11. Conclusion

We demonstrated that LiDAR-based spectral wave estimation can be directly exploited for feedforward ride control of hydrofoil vessels. The key findings are:

1. Wave-preview feedforward reduces heave acceleration by 11–41% with perfect wave knowledge, and 11–38% with WaveSplat estimation ($\sim$20% nRMSE), on top of an already-effective PD feedback baseline.
2. The feedforward is robust to estimation error: no degradation threshold was found up to 50% nRMSE, thanks to IIR low-pass filtering (α=0.05). This means that wave estimation systems do not need to achieve very high accuracy to provide meaningful ride quality improvement.
3. Optimal preview is short (0.1–0.3 s), matching the spatial scale of wave-foil interaction rather than vessel-scale preview. This is advantageous because LiDAR accuracy is highest in the near field where the preview points are located.
4. End-to-end, WaveSplat feedforward achieves 92–96% of perfect-knowledge performance in BF3, improving ISO 2631 comfort from "Slight discomfort" to "Comfortable"—a practically meaningful improvement for commercial passenger operations.

5. Directional estimation accuracy is more critical than amplitude accuracy for feedforward performance, as evidenced by the BF4 degradation (WS/GT = 37%). This finding provides a clear design target for wave estimation system improvement.

These results establish that even moderate wave estimation accuracy ($\sim$20% nRMSE) is sufficient for effective feedforward control. The total computational budget for the LiDAR $\rightarrow$ WaveSplat $\rightarrow$ feedforward pipeline is 14.6 ms per cycle (Table 6: 13.4 ms estimation + 1.2 ms preprocessing, controller, and I/O), well within the 100 ms real-time budget on standard shipboard computing hardware. This makes the approach a practical candidate for deployment on the emerging class of electric hydrofoil passenger vessels, where ride comfort directly impacts commercial viability.

Roadmap for future work. The natural next steps are: (1) extending the dynamics model to 6-DOF with roll, which enables beam-sea roll control via differential port/starboard foil actuation; (2) replacing the proportional feedforward with model-predictive control (MPC) that can handle actuator constraints ($\pm 15^\circ$ flap limits) and optimize over a receding horizon, potentially improving performance in rough seas where actuator saturation is a concern; (3) longer simulation campaigns (60–120 s) with multiple random seeds for statistically robust comfort assessment; (4) directional confidence weighting in the feedforward gain, reducing the feedforward authority for components with uncertain propagation direction; and (5) validation on a physical hydrofoil testbed to confirm that the simulated benefits translate to measurable comfort improvement under real sea conditions, sensor noise, and actuator dynamics.

CRediT authorship contribution statement

Saeyong Park: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code of the hydrofoil simulator, the feedforward controller, the WaveSplat bridge, and all experiment/figure generation scripts are openly available in the project repository at `https://github.com/SAEYONGPARK94/wavesplat`. Five representative LiDAR episodes (Beaufort 2–5) used for the end-to-end demonstration are included in the repository; the full 100-episode dataset can be regenerated from the released WaveSim simulator with the provided configuration files and fixed random seeds.

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